

CISCO IOS-XE LAB MANUAL

Site-to-Site VPN: IPsec over GRE

Crypto Map | IPsec Profile | VTI | IKEv1 | IKEv2

Platform Coverage: Cisco IOS-XE · ASA 5500-X · Firepower FTD

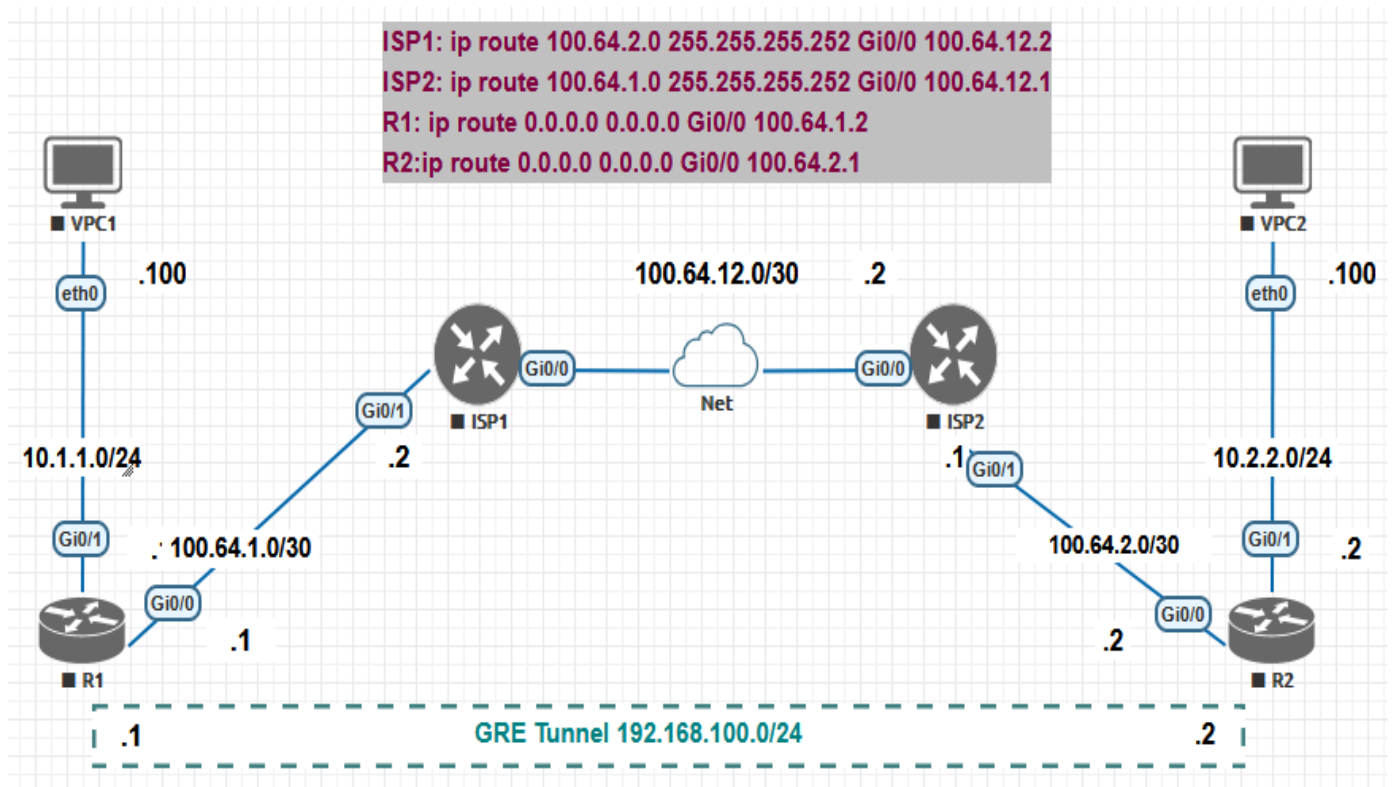


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1. Lab Overview

1.1 Objectives

- Understand GRE tunnel fundamentals and why GRE is used as a transport for IPsec
- Configure IPsec over GRE using the legacy Crypto Map method (IKEv1 / IKEv2)
- Configure IPsec over GRE using the modern IPsec Profile method
- Configure Site-to-Site VPN using Virtual Tunnel Interface (VTI) over IPsec
- Apply encryption, HMAC integrity, and peer authentication algorithms supported by IOS-XE, ASA 5500-X, and Firepower FTD
- Verify, troubleshoot, and validate all VPN scenarios end-to-end

1.2 Topology Summary

The lab uses five devices. VPC1 and VPC2 are end hosts on private RFC-1918 subnets. R1 and R2 are the CE (Customer Edge) routers that will terminate all VPN configurations. ISP1 and ISP2 simulate the public Internet with a /30 interconnect.

Device	Interface	IP Address	Subnet / Role
VPC1	eth0	10.1.1.100	10.1.1.0/24 – LAN A
R1	Gi0/1	10.1.1.1	10.1.1.0/24 – LAN A gateway
R1	Gi0/0	100.64.1.1	100.64.1.0/30 – WAN toward ISP1
ISP1	Gi0/1	100.64.1.2	100.64.1.0/30
ISP1	Gi0/0	100.64.12.1	100.64.12.0/30 – ISP Interconnect
ISP2	Gi0/0	100.64.12.2	100.64.12.0/30
ISP2	Gi0/1	100.64.2.1	100.64.2.0/30 – WAN toward R2
R2	Gi0/0	100.64.2.2	100.64.2.0/30 – WAN toward ISP2
R2	Gi0/1	10.2.2.2	10.2.2.0/24 – LAN B
VPC2	eth0	10.2.2.100	10.2.2.0/24 – LAN B
GRE Tunnel	Tunnel0 (R1)	192.168.100.1	192.168.100.0/24
GRE Tunnel	Tunnel0 (R2)	192.168.100.2	192.168.100.0/24

1.3 Routing – Pre-configured Static Routes

The following static routes must be in place before any VPN configuration begins:

```
! R1 - default route to ISP1
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0 100.64.1.2

! R2 - default route to ISP2
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0 100.64.2.1

! ISP1 - static routes for reachability
ip route 100.64.2.0 255.255.255.252 GigabitEthernet0/0 100.64.12.2
```

```
! ISP2 - static routes for reachability
ip route 100.64.1.0 255.255.255.252 GigabitEthernet0/0 100.64.12.1
```

i NOTE: Static Routes Only

No routing protocols are pre-configured. The static default routes ensure R1 and R2 can reach each other's WAN IPs across the simulated Internet. Dynamic routing (e.g., OSPF over GRE) can be added on top of the VPN in later labs.

1.4 Algorithm Support Matrix

All algorithms below are supported on IOS-XE, ASA 5500-X (running ASA OS 9.x), and Firepower FTD. Algorithms marked with **X** are deprecated or unsupported on one or more platforms.

Category	Algorithm	IOS-XE	ASA 5500-X	FTD	Notes
Encryption	AES-128-CBC	✓	✓	✓	Minimum recommended
Encryption	AES-256-CBC	✓	✓	✓	Recommended
Encryption	AES-256-GCM	✓	✓	✓	IKEv2 / Suite-B only
Encryption	3DES	✓	✓	✓	Legacy – avoid in new designs
Encryption	DES	✓	X	X	Prohibited – RFC 6151
HMAC/Integrity	SHA-256 (HMAC)	✓	✓	✓	Recommended
HMAC/Integrity	SHA-384	✓	✓	✓	Suite-B
HMAC/Integrity	SHA-512	✓	✓	✓	Suite-B
HMAC/Integrity	SHA-1 (HMAC)	✓	✓	✓	Legacy – avoid
HMAC/Integrity	MD5	✓	✓	✓	Deprecated
IKE Auth	Pre-Shared Key	✓	✓	✓	Simple – used in this lab
IKE Auth	RSA Signature	✓	✓	✓	PKI – enterprise best practice
IKE Auth	ECDSA	✓	✓	✓	Suite-B
DH Group	Group 14 (2048)	✓	✓	✓	Minimum recommended
DH Group	Group 19 (256-ECP)	✓	✓	✓	Suite-B
DH Group	Group 20 (384-ECP)	✓	✓	✓	Suite-B
DH Group	Group 21 (521-ECP)	✓	✓	✓	Suite-B
DH Group	Group 5 (1536)	✓	✓	✓	Legacy – avoid
DH Group	Group 1/2	✓	X	X	Prohibited

1.5 Lab Sections

1. Prerequisites and IP Address Verification
2. Part A – GRE Tunnel (No Encryption)
3. Part B – IPsec over GRE using Crypto Map (IKEv1)
4. Part C – IPsec over GRE using Crypto Map (IKEv2)
5. Part D – IPsec over GRE using IPsec Profile
6. Part E – Site-to-Site VTI over IPsec
7. Verification and Troubleshooting Reference
8. Review Questions and Answers

2. Prerequisites and IP Verification

2.1 Configure Base Interfaces

R1 Base Configuration

```
hostname R1
!
interface GigabitEthernet0/0
description WAN-to-ISP1
ip address 100.64.1.1 255.255.255.252
no shutdown
!
interface GigabitEthernet0/1
description LAN-A
ip address 10.1.1.1 255.255.255.0
no shutdown
!
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0 100.64.1.2
```

R2 Base Configuration

```
hostname R2
!
interface GigabitEthernet0/0
description WAN-to-ISP2
ip address 100.64.2.2 255.255.255.252
no shutdown
!
interface GigabitEthernet0/1
description LAN-B
ip address 10.2.2.2 255.255.255.0
no shutdown
!
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0 100.64.2.1
```

ISP1 and ISP2 Base Configuration

```
! ISP1
hostname ISP1
interface GigabitEthernet0/0
description LINK-TO-ISP2
ip address 100.64.12.1 255.255.255.252
```

```

no shutdown
interface GigabitEthernet0/1
description LINK-TO-R1
ip address 100.64.1.2 255.255.255.252
no shutdown
exit
ip route 100.64.2.0 255.255.255.252 GigabitEthernet0/0 100.64.12.2
!
! ISP2
hostname ISP2
interface GigabitEthernet0/0
description LINK-TO-ISP1
ip address 100.64.12.2 255.255.255.252
no shutdown
interface GigabitEthernet0/1
description LINK-TO-R2
ip address 100.64.2.1 255.255.255.252
no shutdown
exit
ip route 100.64.1.0 255.255.255.252 GigabitEthernet0/0 100.64.12.1

```

2.2 Verification – Underlay Reachability

Confirm WAN reachability between R1 and R2 before building any tunnel:

```

! From R1 - ping R2's WAN IP
R1# ping 100.64.2.2 source GigabitEthernet0/0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

! From R2 - ping R1's WAN IP
R2# ping 100.64.1.1 source GigabitEthernet0/0
!!!!

```

✗ CRITICAL: Do Not Proceed Without 100% Ping Success

All subsequent VPN configurations depend on WAN reachability. If pings fail, re-verify ISP static routes and interface states before continuing.

3. Part A – GRE Tunnel (No Encryption)

3.1 Concept

Generic Routing Encapsulation (GRE) is a tunneling protocol defined in RFC 2784 that encapsulates any Layer-3 protocol inside an IP packet. GRE itself provides no encryption, authentication, or integrity protection – it is simply a mechanism to extend a routed network across a transit (underlay) network.

GRE is used as a carrier for IPsec in this lab because:

- GRE supports multicast and broadcast traffic (required for routing protocols like OSPF or EIGRP over the tunnel)
- IPsec in transport mode can then protect the GRE packet rather than individual flows
- The combination (GRE+IPsec) provides both protocol flexibility and strong encryption

NOTE: GRE Packet Overhead

GRE adds a 4-byte GRE header plus a 20-byte outer IP header. Total overhead = 24 bytes per packet.

With IPsec (ESP) in transport mode, additional overhead is approximately 50-60 bytes.

Ensure MTU is set appropriately to avoid fragmentation (typically 'ip mtu 1400' on the tunnel interface).

3.2 GRE Configuration – R1

```
interface Tunnel0
description GRE-Tunnel-to-R2
ip address 192.168.100.1 255.255.255.0
tunnel source GigabitEthernet0/0
tunnel destination 100.64.2.2
ip mtu 1400
ip tcp adjust-mss 1360
no shutdown
```

3.3 GRE Configuration – R2

```
interface Tunnel0
description GRE-Tunnel-to-R1
ip address 192.168.100.2 255.255.255.0
tunnel source GigabitEthernet0/0
tunnel destination 100.64.1.1
ip mtu 1400
ip tcp adjust-mss 1360
no shutdown
```

3.4 Static Routes over GRE (Optional – for plain GRE test)

```
! R1 - reach LAN-B via tunnel
ip route 10.2.2.0 255.255.255.0 Tunnel0

! R2 - reach LAN-A via tunnel
ip route 10.1.1.0 255.255.255.0 Tunnel0
```

3.5 GRE Verification

```
! Check tunnel interface state
R1# show interfaces Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 192.168.100.1/24
  Tunnel source 100.64.1.1 (GigabitEthernet0/0), destination 100.64.2.2
```

```
Tunnel protocol/transport GRE/IP
```

```
! Ping tunnel peer
```

```
R1# ping 192.168.100.2
```

```
!!!!!
```

```
! Ping remote LAN
```

```
R1# ping 10.2.2.100 source 10.1.1.1
```

```
!!!!!
```

✓ TIP: Tunnel0 Line Protocol

Tunnel0 line protocol comes up when GRE keepalives are received (if configured) OR when the tunnel destination is reachable via the routing table. Without keepalives, the tunnel can remain 'up/up' even if the remote end is misconfigured. Add 'keepalive 10 3' for better failure detection.

4. Part B – IPSec over GRE using Crypto Map (IKEv1)

4.1 Concept – Crypto Map Method

The Crypto Map method is the traditional (legacy) way to configure IPSec on Cisco IOS. It involves three building blocks:

- **ISAKMP Policy (Phase 1 / IKE SA)** – Negotiates the management tunnel for key exchange
- **IPSec Transform Set (Phase 2 / IPSec SA)** – Defines encryption and integrity for data traffic
- **Crypto Map** – Ties together: the match ACL (interesting traffic), peer IP, and transform set, then applies to an interface

In GRE+IPSec, the 'interesting traffic' for IPSec is not the LAN subnets – it is the GRE protocol traffic between the two WAN IPs. IPSec encrypts the outer GRE packet. LAN traffic rides inside the (already-encrypted) GRE tunnel.

⚠ WARNING: ACL Scope for GRE+IPSec Crypto Map

The crypto ACL must match GRE protocol (IP protocol 47) between the tunnel endpoints, NOT the LAN subnets.

Matching LAN-to-LAN traffic directly with a crypto map (without GRE) creates a crypto map tunnel mode VPN – a different design covered separately.

4.2 IKEv1 Phase 1 – ISAKMP Policy

R1 Configuration

```
! Step 1: Define ISAKMP (IKEv1) Phase-1 policy
```

```
crypto isakmp policy 10
```

```
  encryption aes 256
```

```
  hash sha256
```

```
  authentication pre-share
```

```
  group 14
```

```
lifetime 86400
!
! Step 2: Define Pre-Shared Key for R2's WAN IP
crypto isakmp key Cisco@12345 address 100.64.2.2
```

R2 Configuration

```
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
 lifetime 86400
!
crypto isakmp key Cisco@12345 address 100.64.1.1
```

i NOTE: ISAKMP Policy Matching

IKEv1 Phase-1 policies are compared in order (lowest number first). Both peers must have at least one matching policy. The lifetime can differ; the shorter value wins.

DH Group 14 (2048-bit) is the minimum recommended. Groups 19/20/21 (ECDH) are preferred for Suite-B compliance.

4.3 IPsec Phase 2 – Transform Set

R1 and R2 (same on both)

```
! Define transform set
! esp-aes 256 = AES-256-CBC encryption
! esp-sha256-hmac = HMAC-SHA-256 integrity
crypto ipsec transform-set GRE-PROTECT esp-aes 256 esp-sha256-hmac
 mode transport
!
! Note: 'mode transport' is used because GRE already provides the outer IP
header.
! IPsec encrypts the GRE payload only (efficient).
! 'mode tunnel' would add yet another IP header (wasteful for GRE-over-IPsec).
```

i NOTE: Transport vs Tunnel Mode for GRE+IPsec

Transport Mode: IPsec encrypts the payload (GRE packet body) and adds ESP header/trailer. The original outer IP header is preserved. Overhead is lower.

Tunnel Mode: IPsec encapsulates the entire GRE packet inside a new IP header. This adds 20 bytes of additional overhead and is typically unnecessary when GRE already provides encapsulation.

Best practice for GRE+IPsec: use transport mode.

4.4 Crypto ACL – Match GRE Traffic

R1

```
! Match GRE (protocol 47) FROM R1-WAN TO R2-WAN
ip access-list extended ACL-GRE-VPN
```

```
permit gre host 100.64.1.1 host 100.64.2.2
```

R2

```
ip access-list extended ACL-GRE-VPN
permit gre host 100.64.2.2 host 100.64.1.1
```

4.5 Crypto Map

R1

```
crypto map VPN-MAP 10 ipsec-isakmp
description IPsec-over-GRE-to-R2
set peer 100.64.2.2
set transform-set GRE-PROTECT
match address ACL-GRE-VPN
set pfs group14
!
! Apply crypto map to WAN interface
interface GigabitEthernet0/0
crypto map VPN-MAP
```

R2

```
crypto map VPN-MAP 10 ipsec-isakmp
description IPsec-over-GRE-to-R1
set peer 100.64.1.1
set transform-set GRE-PROTECT
match address ACL-GRE-VPN
set pfs group14
!
interface GigabitEthernet0/0
crypto map VPN-MAP
```

4.6 Static Routes over Tunnel (after IPsec)

```
! R1
ip route 10.2.2.0 255.255.255.0 Tunnel0

! R2
ip route 10.1.1.0 255.255.255.0 Tunnel0
```

4.7 Phase 1 & Phase 2 Verification

```
! Trigger tunnel - ping from LAN to LAN
R1# ping 10.2.2.100 source 10.1.1.1

! Verify IKEv1 Phase 1 SA
R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
100.64.2.2   100.64.1.1   QM_IDLE       1001 ACTIVE
```

```

! Verify IPsec Phase 2 SA
R1# show crypto ipsec sa
interface: GigabitEthernet0/0
  Crypto map tag: VPN-MAP, local addr 100.64.1.1
  protected vrf: (none)
  local ident (addr/mask/prot/port): (100.64.1.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (100.64.2.2/255.255.255.255/47/0)
  #pkts encaps: 15, #pkts encrypt: 15, #pkts digest: 15
  #pkts decaps: 15, #pkts decrypt: 15, #pkts verify: 15
  inbound esp sas:
    spi: 0xABCD1234 (2882400820)
    transform: esp-256-aes esp-sha256-hmac
    in use settings ={Transport, }
  outbound esp sas:
    spi: 0x1234ABCD (305441229)
    transform: esp-256-aes esp-sha256-hmac
    in use settings ={Transport, }

! Check encrypt/decrypt counters increase with traffic
R1# show crypto ipsec sa | include pkts

```

5. Part C – IPsec over GRE using Crypto Map (IKEv2)

5.1 IKEv2 Overview

IKEv2 (RFC 7296) supersedes IKEv1 and provides:

- Fewer message exchanges (4 messages vs 9 in IKEv1 main mode) – faster SA establishment
- Built-in NAT-T (NAT Traversal) support
- Asymmetric authentication (each peer can use a different auth method)
- Improved resistance to DoS attacks with cookies
- MOBIKE support for mobile clients (RFC 4555)

IKEv2 replaces the concept of 'ISAKMP policy' with a set of IKEv2-specific constructs: proposal, policy, profile, and keyring.

✗ CRITICAL: IKEv2 with crypto map requires IOS-XE 16.x or later (CSR1000v works)— 15.8(3)M2 vIOS supports IKEv2 configuration syntax but has a known initiator bug that prevents SA establishment.

5.2 IKEv2 Proposal (algorithms)

```

! R1 and R2 - IKEv2 Proposal (Phase 1 algorithms)
crypto ikev2 proposal IKEv2-PROP
  encryption aes-cbc-256
  integrity sha256
  group 14
!
! For Suite-B (ECDH + AES-GCM), use:

```

```
! encryption aes-gcm-256
! integrity sha384
! group 20
```

5.3 IKEv2 Policy

```
crypto ikev2 policy IKEv2-POL
proposal IKEv2-PROP
```

5.4 IKEv2 Keyring (Pre-Shared Keys)

R1

```
crypto ikev2 keyring IKEv2-KEYS
peer R2
address 100.64.2.2
pre-shared-key Cisco@IKEv2#R1R2
```

R2

```
crypto ikev2 keyring IKEv2-KEYS
peer R1
address 100.64.1.1
pre-shared-key Cisco@IKEv2#R1R2
```

5.5 IKEv2 Profile

R1

```
crypto ikev2 profile IKEv2-PROF
match identity remote address 100.64.2.2 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local IKEv2-KEYS
lifetime 86400
```

R2

```
crypto ikev2 profile IKEv2-PROF
match identity remote address 100.64.1.1 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local IKEv2-KEYS
lifetime 86400
```

5.6 Transform Set and Crypto Map with IKEv2

The transform set is the same as Part B. The crypto map references the IKEv2 profile:

R1 and R2

```
! Transform set (same as IKEv1 Part B)
crypto ipsec transform-set GRE-PROTECT-V2 esp-aes 256 esp-sha256-hmac
```

```

mode transport
!
! Crypto Map - adds ikev2-profile reference
crypto map VPN-MAP-V2 10 ipsec-isakmp
  set peer 100.64.2.2          ! (100.64.1.1 on R2)
  set transform-set GRE-PROTECT-V2
  set ikev2-profile IKEv2-PROF
  match address ACL-GRE-VPN
  set pfs group14
!
interface GigabitEthernet0/0
  crypto map VPN-MAP-V2

```

NOTE: Removing IKEv1 Config

If switching from IKEv1 to IKEv2 on an existing router, remove the old 'crypto isakmp policy' and 'crypto isakmp key' entries. IKEv1 and IKEv2 can coexist but may cause confusion during troubleshooting.

To disable IKEv1: 'no crypto isakmp enable' (use with caution in production).

5.7 IKEv2 Verification

```

! IKEv2 SA status
R1# show crypto ikev2 sa
  IPv4 Crypto IKEv2  SA

Tunnel-id Local Remote fvrf/ivrf Status
1 100.64.1.1/500 100.64.2.2/500 none/none READY
  Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth
  sign: PSK, Auth verify: PSK
  Life/Active Time: 86400/342 sec

! IKEv2 detailed statistics
R1# show crypto ikev2 stats

! Verify Phase 2 (same as IKEv1)
R1# show crypto ipsec sa

```

6. Part D – IPsec over GRE using IPsec Profile

6.1 Concept – IPsec Profile Method

The IPsec Profile method (also called the 'tunnel protection' approach) binds IPsec directly to a tunnel interface rather than to a physical WAN interface via a crypto map. This is the preferred, modern approach for GRE+IPsec.

Advantages over Crypto Map:

- No crypto ACL required – any traffic routed into the tunnel interface is automatically protected

- Cleaner configuration – crypto is attached to the logical tunnel, not the physical WAN
- Mandatory for dynamic (SVTI/DVTI) tunnel configurations
- Works with both IKEv1 and IKEv2

✓ TIP: Tunnel Protection vs Crypto Map

When using 'tunnel protection ipsec profile', remove any 'crypto map' from Gi0/0. The two methods are mutually exclusive on the same interface pair.

6.2 ISAKMP/IKE Phase 1 – same as Parts B or C

Reuse the IKEv1 ISAKMP policy or IKEv2 profile from Part B or C. No changes required for Phase 1 when switching from crypto map to IPsec profile.

6.3 IPsec Transform Set

```
! R1 and R2
crypto ipsec transform-set TS-GRE-PROFILE esp-aes 256 esp-sha256-hmac
mode transport
```

6.4 IPsec Profile

```
! R1 and R2 - using IKEv1 ISAKMP
crypto ipsec profile IPSEC-GRE-PROFILE
set transform-set TS-GRE-PROFILE
set pfs group14
!
! Optional: set security-association lifetime seconds 3600
! Optional: set security-association replay window-size 512
```

! NOTE: Using IKEv2 Profile with IPsec Profile

To use IKEv2 instead of IKEv1 with IPsec profile, add:

```
crypto ipsec profile IPSEC-GRE-PROFILE
set ikev2-profile IKEv2-PROF
set transform-set TS-GRE-PROFILE
```

Both IKEv1 and IKEv2 can be referenced in the same profile; IKEv2 is attempted first.

6.5 Tunnel Interface with Tunnel Protection

R1

```
interface Tunnel0
description GRE-IPsec-Profile-to-R2
ip address 192.168.100.1 255.255.255.0
tunnel source GigabitEthernet0/0
tunnel destination 100.64.2.2
tunnel mode gre ip
tunnel protection ipsec profile IPSEC-GRE-PROFILE
ip mtu 1400
ip tcp adjust-mss 1360
```



```
no shutdown
```

R2

```
interface Tunnel0
  description GRE-IPSec-Profile-to-R1
  ip address 192.168.100.2 255.255.255.0
  tunnel source GigabitEthernet0/0
  tunnel destination 100.64.1.1
  tunnel mode gre ip
  tunnel protection ipsec profile IPSEC-GRE-PROFILE
  ip mtu 1400
  ip tcp adjust-mss 1360
  no shutdown
```

6.6 Static Routes (same as previous parts)

```
! R1
ip route 10.2.2.0 255.255.255.0 Tunnel0

! R2
ip route 10.1.1.0 255.255.255.0 Tunnel0
```

6.7 Verification

```
! Tunnel state
R1# show interfaces Tunnel0 | include line protocol|protect
Tunnel0 is up, line protocol is up
  Tunnel protection via IPsec (profile "IPSEC-GRE-PROFILE", Mixed-mode)

! IPsec SA (transport mode on tunnel interface)
R1# show crypto ipsec sa interface Tunnel0

! Confirm traffic encrypted
R1# ping 10.2.2.100 source 10.1.1.1 repeat 50
R1# show crypto ipsec sa | include pkts encaps|pkts decaps
      #pkts encaps: 55, #pkts encrypt: 55, #pkts digest: 55
      #pkts decaps: 55, #pkts decrypt: 55, #pkts verify: 55
```

7. Part E – Site-to-Site VTI over IPsec

7.1 Concept – Virtual Tunnel Interface (VTI)

A Virtual Tunnel Interface (VTI) is a routed interface that provides a point-to-point IPsec tunnel without the need for GRE encapsulation. VTI uses IPsec tunnel mode natively on the tunnel interface itself.

Key differences between VTI and GRE+IPsec:

Feature	GRE + IPsec	IPsec VTI
Encapsulation overhead	~24-60 bytes (GRE + ESP)	~52 bytes (ESP tunnel mode only)
Multicast / Routing protocols	Yes (GRE carries multicast)	Yes (VTI supports multicast natively on IOS-XE)
QoS on tunnel	Limited	Full QoS on virtual interface
Configuration complexity	Higher (GRE + IPsec separate)	Lower (single tunnel interface)
Dynamic tunnels (DVTI)	Not supported	Supported
NAT-T support	Requires additional config	Built-in with IKEv2
Platform support	IOS-XE, ASA, FTD	IOS-XE native; ASA/FTD use s2s VPN policies

7.2 Static VTI (S-VTI) Configuration

Phase 1 – IKEv2 Policy (recommended for VTI)

```
! R1 and R2 - IKEv2 Proposal
crypto ikev2 proposal VTI-PROP
  encryption aes-cbc-256
  integrity sha256
  group 14
!
crypto ikev2 policy VTI-POL
  proposal VTI-PROP
```

IKEv2 Keyring

```
! R1
crypto ikev2 keyring VTI-KEYS
  peer R2
    address 100.64.2.2
    pre-shared-key VTI@SecretKey#2024
!
! R2
crypto ikev2 keyring VTI-KEYS
  peer R1
    address 100.64.1.1
    pre-shared-key VTI@SecretKey#2024
```

IKEv2 Profile

```
! R1
crypto ikev2 profile VTI-PROF
  match identity remote address 100.64.2.2 255.255.255.255
  authentication remote pre-share
  authentication local pre-share
  keyring local VTI-KEYS
!
```

```

! R2
crypto ikev2 profile VTI-PROF
match identity remote address 100.64.1.1 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local VTI-KEYS

```

Phase 2 – IPsec Profile for VTI

```

! R1 and R2
crypto ipsec transform-set VTI-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
! NOTE: VTI uses tunnel mode (not transport mode)
! The VTI interface itself is the IP endpoint - tunnel mode is required.
!
crypto ipsec profile VTI-IPSEC-PROFILE
set transform-set VTI-TS
set ikev2-profile VTI-PROF
set pfs group14

```

✗ CRITICAL: Tunnel Mode Required for VTI

Unlike GRE+IPsec (which uses transport mode), VTI must use IPsec tunnel mode. The VTI tunnel interface is a virtual IP endpoint; IPsec must encapsulate the entire inner IP packet with a new outer IP header. Using transport mode with VTI will cause the tunnel to fail to pass traffic.

VTI Tunnel Interface – R1

```

interface Tunnel10
description S-VTI-IPsec-to-R2
ip address 192.168.200.1 255.255.255.0
tunnel source GigabitEthernet0/0
tunnel destination 100.64.2.2
tunnel mode ipsec ipv4
tunnel protection ipsec profile VTI-IPSEC-PROFILE
ip mtu 1438
ip tcp adjust-mss 1398
no shutdown

```

VTI Tunnel Interface – R2

```

interface Tunnel10
description S-VTI-IPsec-to-R1
ip address 192.168.200.2 255.255.255.0
tunnel source GigabitEthernet0/0
tunnel destination 100.64.1.1
tunnel mode ipsec ipv4
tunnel protection ipsec profile VTI-IPSEC-PROFILE
ip mtu 1438
ip tcp adjust-mss 1398
no shutdown

```

NOTE: VTI MTU Calculation

IPsec ESP tunnel mode overhead: 20 (outer IP) + 8 (ESP) + 16 (IV) + 16 (ESP auth) + up to 15 (padding) = ~59 bytes

Effective payload MTU = 1500 - 59 = 1441. Use 1438 for alignment safety.

MSS = MTU - 40 (IP + TCP headers) = 1398.

Static Routes over VTI

```
! R1
ip route 10.2.2.0 255.255.255.0 Tunnel10

! R2
ip route 10.1.1.0 255.255.255.0 Tunnel10
```

7.3 VTI Verification

```
! Verify VTI tunnel mode
R1# show interfaces Tunnel10
Tunnel10 is up, line protocol is up
  Tunnel source 100.64.1.1 (GigabitEthernet0/0), destination 100.64.2.2
  Tunnel protocol/transport IPSEC/IP
  Tunnel protection via IPsec (profile "VTI-IPSEC-PROFILE", Mixed-mode)

! IKEv2 SA
R1# show crypto ikev2 sa

! IPsec SA - note tunnel mode
R1# show crypto ipsec sa interface Tunnel10
  in use settings ={Tunnel, }

! End-to-end test
R1# ping 10.2.2.100 source 10.1.1.1
!!!!

! Verify packet counts
R1# show crypto ipsec sa | include pkts
```

7.4 Routing Protocols over VTI

OSPF can run directly over the VTI interface – no GRE encapsulation needed:

```
! R1 - OSPF over VTI
router ospf 1
  router-id 1.1.1.1
  network 192.168.200.0 0.0.0.255 area 0
  network 10.1.1.0 0.0.0.255 area 0
!

! R2 - OSPF over VTI
router ospf 1
  router-id 2.2.2.2
```

```
network 192.168.200.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
```

8. Verification and Troubleshooting Reference

8.1 Master Verification Checklist

Step	Command	Expected Output
1	show interfaces Tunnel0	Line protocol: up
2	ping 192.168.100.2 (GRE+IPsec)	5/5 success
3	show crypto isakmp sa	QM_IDLE or MM_ACTIVE
4	show crypto ikev2 sa	READY
5	show crypto ipsec sa	#pkts encaps/decaps > 0
6	show crypto session	Session status: UP-ACTIVE
7	ping 10.2.2.100 source 10.1.1.1	5/5 success
8	show crypto ipsec sa incl pkts	Counters incrementing

8.2 Debug Commands (use with caution in production)

```
! IKEv1 Phase 1 negotiation
debug crypto isakmp

! IKEv2 negotiation
debug crypto ikev2

! IPsec SA negotiation (Phase 2)
debug crypto ipsec

! GRE tunnel events
debug tunnel

! Always disable debugs after use
undebug all

! OR
no debug all
```

⚠ WARNING: Debug in Production

Crypto debugs are CPU-intensive and can generate massive log output on busy routers. Always use 'condition' debugs to filter by peer IP in production:

```
debug crypto condition peer ipv4 100.64.2.2
```

Clear condition after troubleshooting: 'no debug condition all'

8.3 Common Issues and Fixes

Symptom	Likely Cause	Fix
show crypto isakmp sa shows MM_NO_STATE	Phase 1 policy mismatch or peer unreachable	Check encryption/hash/DH group match; verify underlay ping
show crypto isakmp sa shows MM_KEY_EXCH	PSK mismatch	Verify 'crypto isakmp key' matches exactly on both peers
show crypto ipsec sa shows 0 pkts encaps	Interesting traffic not matching ACL	Check crypto ACL – must match GRE (proto 47) not LAN subnets
Tunnel0 up/down (line protocol down)	No GRE path (routing issue)	Verify ip route for tunnel destination; check underlay
Ping across tunnel fails	Missing static route over tunnel	Add 'ip route 10.x.x.x 255.x.x.x Tunnel0'
IKEv2 show crypto ikev2 sa empty	IKEv2 profile not matching peer identity	Check 'match identity remote address' statement
ESP packets visible in wireshark unencrypted	Transform set mode is wrong (tunnel instead of transport on GRE)	Set 'mode transport' in transform-set for GRE+IPsec
VTI Tunnel10 flaps up/down repeatedly	IKEv2 DPD (dead peer detection) firing due to no traffic	Add 'set dpd 30 2 periodic' in ikev2 profile

8.4 Clearing IPsec and IKE SAs

```

! Clear all IKEv1 SAs (triggers re-negotiation)
clear crypto isakmp

! Clear IKEv1 SA for specific peer
clear crypto isakmp 100.64.2.2

! Clear all IPsec SAs
clear crypto sa

! Clear IPsec SA for specific peer
clear crypto sa peer 100.64.2.2

! Clear IKEv2 SA
clear crypto ikev2 sa

```

8.5 show crypto session – Comprehensive Status

```

R1# show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection

```

```

K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: GigabitEthernet0/0
Profile: IKEv2-PROF
Uptime: 00:12:34
Session status: UP-ACTIVE
Peer: 100.64.2.2 port 500 fvrf: (none) ivrf: (none)
  Phase1_id: 100.64.2.2
  Desc: (none)
Session ID: 1
IKEv2 SA: local 100.64.1.1/500 remote 100.64.2.2/500 Active
  Capabilities:D connid:1 lifetime:23:47:26
IPSEC FLOW: permit 47 host 100.64.1.1 host 100.64.2.2
  Active SAs: 2, origin: crypto map

```

9. ASA 5500-X and Firepower FTD – Configuration Equivalents

9.1 ASA 5500-X (CLI) – Site-to-Site IPsec IKEv2

The ASA does not support GRE tunnels natively. Site-to-site VPN on ASA uses crypto maps with LAN-to-LAN interesting traffic (tunnel mode). The following mirrors the same encryption parameters.

```

! ASA - IKEv2 Policy
crypto ikev2 policy 10
  encryption aes-256
  integrity sha256
  group 14
  prf sha256
  lifetime seconds 86400
!
! ASA - IKEv2 Enable on outside interface
crypto ikev2 enable outside
!
! ASA - Tunnel Group (replaces IOS keyring)
tunnel-group 100.64.2.2 type ipsec-l2l
tunnel-group 100.64.2.2 ipsec-attributes
  ikev2 remote-authentication pre-shared-key Cisco@IKEv2#R1R2
  ikev2 local-authentication pre-shared-key Cisco@IKEv2#R1R2
!
! ASA - IKEv2 IPsec Proposal (replaces transform-set)
crypto ipsec ikev2 ipsec-proposal ASA-PROP
  protocol esp encryption aes-256
  protocol esp integrity sha-256
!
! ASA - Crypto ACL (LAN-to-LAN, tunnel mode on ASA)
access-list VPN-ACL extended permit ip 10.1.1.0 255.255.255.0 10.2.2.0
255.255.255.0

```

```

!
! ASA - Crypto Map
crypto map OUTSIDE-MAP 10 match address VPN-ACL
crypto map OUTSIDE-MAP 10 set peer 100.64.2.2
crypto map OUTSIDE-MAP 10 set ikev2 ipsec-proposal ASA-PROP
crypto map OUTSIDE-MAP 10 set pfs group14
crypto map OUTSIDE-MAP interface outside
!
! ASA - NAT Exemption (required to bypass NAT for VPN traffic)
nat (inside,outside) source static LAN-A LAN-A destination static LAN-B LAN-B

```

i NOTE: ASA IKEv2 vs IKEv1 Syntax

ASA uses 'crypto ikev2 policy' (not 'crypto isakmp policy').

ASA uses 'crypto ipsec ikev2 ipsec-proposal' (not 'crypto ipsec transform-set') for IKEv2.

For IKEv1 on ASA: 'crypto isakmp policy' + 'crypto ipsec transform-set' (same as IOS).

FTD managed via FMC uses GUI policies – no direct CLI IPsec configuration.

9.2 ASA Verification Commands

```

! Phase 1 IKEv2 SA
ASA# show crypto ikev2 sa

! Phase 2 IPsec SA
ASA# show crypto ipsec sa

! VPN session summary
ASA# show vpn-sessiondb 121

! Detailed session
ASA# show vpn-sessiondb detail 121

! Debug (use sparingly)
ASA# debug crypto ikev2 protocol 127
ASA# debug crypto ikev2 platform 127

```

9.3 Firepower FTD – GUI-Based S2S VPN

FTD managed by Firepower Management Center (FMC) uses a GUI wizard for Site-to-Site VPN. The equivalent parameters are:

FMC Field	This Lab Value
VPN Type	Point-to-Point
IKE Version	IKEv2
Encryption Algorithm	AES-256
Hash Algorithm	SHA-256
DH Group	Group 14
Authentication Method	Pre-Shared Key

FMC Field	This Lab Value
ESP Encryption	AES-256
ESP Integrity	SHA-256
PFS DH Group	Group 14
Local Networks	10.1.1.0/24
Remote Networks	10.2.2.0/24

✓ TIP: FTD NAT Exemption

In FMC, create an Identity NAT rule: Source Zone = inside, Destination Zone = outside, Source Network = LAN-A, Destination Network = LAN-B, Action = Do Not Translate.
Without NAT exemption, VPN traffic will be NATted before matching the VPN policy and the tunnel will pass no traffic.

10. Review Questions and Answers

10.1 Multiple Choice

Q1. Which protocol number does GRE use in the IP header?

- A) 50
- B) 51
- C) 47
- D) 17

✓ TIP: Answer: C – Protocol 47

GRE uses IP protocol number 47. Protocol 50 = ESP (IPsec Encapsulating Security Payload), Protocol 51 = AH (Authentication Header), Protocol 17 = UDP.
This is why the crypto ACL for GRE+IPsec uses 'permit gre host X host Y' rather than UDP or TCP.

Q2. In a GRE+IPsec configuration, why should the transform set use 'mode transport' rather than 'mode tunnel'?

- A) Tunnel mode is not supported on GRE interfaces
- B) GRE already provides the outer IP header; transport mode avoids adding a redundant second outer IP header
- C) Transport mode provides stronger encryption than tunnel mode
- D) Tunnel mode is only available with IKEv2

✓ TIP: Answer: B

GRE adds its own outer IP header (with source/destination = WAN IPs). IPsec transport mode encrypts only the payload (the GRE packet body), preserving the outer IP header. If tunnel mode were used, a third IP header would be added unnecessarily, wasting bandwidth.
By contrast, VTI (Virtual Tunnel Interface) must use tunnel mode because there is no GRE encapsulation – the VTI itself is the IP endpoint.

Q3. What is the purpose of the 'ip tcp adjust-mss 1360' command on a GRE tunnel interface?

- A) It sets the maximum segment size for ICMP packets
- B) It adjusts TCP SYN packets so that the negotiated MSS prevents IP fragmentation through the tunnel
- C) It limits the tunnel to 1360 kbps throughput
- D) It enables TCP keepalives on the tunnel interface

✓ TIP: Answer: B

When TCP sessions are established through a GRE+IPsec tunnel, the two endpoints negotiate an MSS (Maximum Segment Size) during the TCP three-way handshake. Without MSS clamping, they may agree on 1460 bytes (standard Ethernet). Adding GRE (24 bytes) + IPsec (up to 60 bytes) overhead can cause packets to exceed 1500 bytes, leading to fragmentation.

'ip tcp adjust-mss 1360' modifies TCP SYN packets transiting the interface to cap the MSS at 1360, ensuring the resulting IP packets stay within MTU after encapsulation.

Q4. Which IKE phase negotiates the encryption keys used to protect actual data traffic?

- A) Phase 1 (ISAKMP SA / IKE_SA_INIT)
- B) Phase 1.5 (Mode Config)
- C) Phase 2 (IPsec SA / IKE_AUTH + CREATE_CHILD_SA)
- D) Phase 3 (ESP negotiation)

✓ TIP: Answer: C

Phase 1 (ISAKMP SA in IKEv1 / IKE_SA in IKEv2) creates a secure, authenticated channel between the two peers – this is a management channel only.

Phase 2 (IPsec SA / Quick Mode in IKEv1 / CREATE_CHILD_SA in IKEv2) negotiates the actual data encryption parameters (transform set / IPsec proposal) and generates the symmetric keys used by ESP to encrypt user data traffic.

There is no IKE 'Phase 3'.

Q5. A network engineer sees the following in 'show crypto isakmp sa': state = MM_KEY_EXCH. What is the most likely cause?

- A) DH group mismatch
- B) Pre-shared key mismatch between peers
- C) The tunnel source interface is down
- D) The crypto ACL does not match any traffic

✓ TIP: Answer: B

MM_KEY_EXCH means IKEv1 Main Mode reached message 5 (key exchange) but stalled – typically due to a PSK mismatch. If the DH group didn't match, the failure would occur earlier (MM_NO_STATE or MM_SA_SETUP). If the source interface were down, there would be no ISAKMP SA entry at all.

A crypto ACL mismatch affects Phase 2 (IPsec SA), not Phase 1 (ISAKMP SA).

10.2 Short Answer

Q6. List three advantages of using the IPsec Profile ('tunnel protection') method over the Crypto Map method for GRE+IPsec.

✓ **TIP: Answer**

1. No crypto ACL required – any traffic entering the tunnel interface is automatically encrypted, eliminating the risk of ACL misconfiguration.
 2. Cleaner separation – IPsec is bound to the logical tunnel interface rather than the physical WAN interface, making the configuration more intuitive and easier to manage.
 3. Mandatory for dynamic tunnels – DVTI (Dynamic VTI) and DMVPN require the tunnel protection approach; crypto maps cannot be used.
- Bonus: Easier to apply per-tunnel QoS and interface-level features (e.g., ip accounting, NetFlow) directly on the tunnel interface.

Q7. What command would you use on a Cisco IOS-XE router to verify that IPsec is encrypting traffic over a GRE tunnel and check the packet counters?

✓ **TIP: Answer**

Primary command: show crypto ipsec sa

To filter for packet counters: show crypto ipsec sa | include pkts

Look for '#pkts encaps', '#pkts encrypt', '#pkts decaps', '#pkts decrypt' counters.

If encaps/encrypt counters are 0, IPsec is not encrypting (check ACL or tunnel state).

If decaps/decrypt counters are 0 but encaps is non-zero, the return path is broken (check remote router's SA).

Additional useful command: show crypto session detail (shows up/down status, SA lifetime, and capabilities in one view).

Q8. Explain why 'mode tunnel' must be used in the IPsec transform set when configuring a VTI, but 'mode transport' is recommended for GRE+IPsec.

✓ **TIP: Answer**

In GRE+IPsec: GRE provides the outer IP encapsulation (source = R1 WAN, destination = R2 WAN). IPsec in transport mode encrypts the GRE payload while preserving the existing outer IP header. No additional IP header is needed.

In VTI: There is no GRE encapsulation layer. The VTI itself is a virtual interface with an IP address, but it does not generate an outer IP header automatically. IPsec in tunnel mode adds a new outer IP header (source = tunnel source, destination = tunnel destination) to carry the encrypted inner packet across the Internet.

Using transport mode on a VTI would fail because there would be no outer IP header for routing the ESP packet across the underlay network.

Q9. What is the function of the 'set pfs group14' command in a crypto map or IPsec profile, and why is Perfect Forward Secrecy important?

✓ **TIP: Answer**

PFS (Perfect Forward Secrecy) instructs each Phase 2 (IPsec SA) re-key to perform a fresh Diffie-Hellman exchange, independent of the Phase 1 keying material.

Without PFS: If an attacker compromises the Phase 1 master key (SKEYID), they could potentially derive all Phase 2 session keys from that single compromise.

With PFS: Each Phase 2 SA uses a unique DH exchange. Compromising one session key does not allow decryption of past or future sessions.
 'group14' specifies the 2048-bit MODP DH group for the PFS exchange. Groups 19/20/21 (ECDH) are stronger alternatives for Suite-B compliance.

Q10. An engineer applies 'tunnel protection ipsec profile IPSEC-PROF' to Tunnel0 but forgets to remove the existing 'crypto map VPN-MAP' from Gi0/0. What is the likely outcome?

✓ **TIP: Answer**

The behavior depends on IOS-XE version, but in most cases:

1. The crypto map on Gi0/0 will attempt to match GRE traffic (per the crypto ACL) and apply IPsec – creating a double-encryption scenario.
2. The tunnel protection on Tunnel0 will also attempt to apply IPsec – but the GRE packet is already encrypted by the crypto map before it enters the tunnel.
3. The result is typically that Phase 1 and Phase 2 negotiate for both, but one set of SAs cannot decrypt the other's traffic, causing all traffic to fail.

Fix: Remove 'crypto map VPN-MAP' from Gi0/0 when using tunnel protection on Tunnel0. The two methods are mutually exclusive for the same traffic flow.

Command: interface GigabitEthernet0/0 / no crypto map VPN-MAP

11. Quick Reference Cheat Sheet

11.1 Configuration Method Comparison

Attribute	Crypto Map IKEv1	Crypto Map IKEv2	IPsec Profile	VTI
IKE Version	IKEv1	IKEv2	IKEv1 or IKEv2	IKEv2 (recommended)
Phase 1 Config	crypto isakmp policy	crypto ikev2 proposal/policy/profile	Same as chosen IKE	crypto ikev2 proposal/policy/profile
Phase 2 Config	crypto ipsec transform-set	crypto ipsec transform-set	crypto ipsec profile	crypto ipsec profile (tunnel mode)
Interesting Traffic	Crypto ACL (GRE)	Crypto ACL (GRE)	All tunnel traffic (no ACL)	All tunnel traffic (no ACL)
Applied To	Physical WAN (Gi0/0)	Physical WAN (Gi0/0)	Tunnel interface	Tunnel interface
ESP Mode	transport	transport	transport (GRE)	tunnel (VTI)
Tunnel Command	tunnel mode gre ip	tunnel mode gre ip	tunnel protection ipsec profile	tunnel mode ipsec ipv4
Routing Protocol Support	Yes (via GRE)	Yes (via GRE)	Yes (via GRE)	Yes (native)

11.2 Key Show Commands Reference

Command	What It Shows
show crypto isakmp sa	IKEv1 Phase 1 SA state (QM_IDLE = active)
show crypto ikev2 sa	IKEv2 SA state (READY = active)
show crypto ipsec sa	Phase 2 SA, packet counters, transform, mode
show crypto session	VPN session up/down status summary
show crypto session detail	Session + capabilities (DPD, NAT-T, etc.)
show interfaces Tunnel0	Tunnel state, source/dest, mode, protection
show crypto map	Crypto map contents (peer, ACL, transform)
show crypto ipsec profile	IPSec profile contents
show crypto ikev2 proposal	IKEv2 proposal algorithms
show crypto isakmp policy	IKEv1 policy algorithms
show crypto engine connections active	Active crypto engine sessions